# EE3621: Digital Signal Processing

Lecture 4: Frequency Response, Magnitude/Phase & Group Delay of LTI Systems
(III B.Tech EEE)

Lecture Notes (40-Minute Plan)

## 1 Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 08:00 (8 mins):** LTI Eigenfunctions: Why complex exponentials are unique. Derivation of $H(e^{j\omega})$.
- **08:00 – 15:00 (7 mins):** Magnitude & Phase responses, Rectangular vs. Polar forms.
- **15:00 – 25:00 (10 mins):** Phase Delay vs. Group Delay. Narrowband input derivation (Carrier vs. Envelope delay).
- **25:00 – 35:00 (10 mins):** First-order IIR Filter Example: Detailed algebraic derivation of Group Delay.
- **35:00 – 40:00 (5 mins):** Checkpoints & Concept Recap.

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## 2 Eigenfunctions & The Frequency Response

Complex exponentials $e^{j\omega n}$ are eigenfunctions of linear time-invariant systems. Let the input be $x[n] = e^{j\omega n}$ for $-\infty < n < \infty$. The output $y[n]$ is:

$$y[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k] = \sum_{k=-\infty}^{\infty} h[k]e^{j\omega(n-k)} = e^{j\omega n} \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k} \tag{1}$$

We define the complex scaling factor (eigenvalue) as the frequency response $H(e^{j\omega})$:

$$y[n] = H(e^{j\omega})e^{j\omega n} \tag{2}$$

where:

$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} h[k]e^{-j\omega k} \tag{3}$$

## 3 Magnitude & Phase Responses

The frequency response can be written in polar form:

$$H(e^{j\omega}) = |H(e^{j\omega})|e^{j\theta(\omega)} \tag{4}$$

where $|H(e^{j\omega})|$ is the magnitude response and $\theta(\omega) = \angle H(e^{j\omega})$ is the phase response (Figure 1). For a sinusoidal input $x[n] = A\cos(\omega_0 n + \phi)$, the LTI output is:

$$y[n] = A|H(e^{j\omega_0})|\cos(\omega_0 n + \phi + \theta(\omega_0)) \tag{5}$$

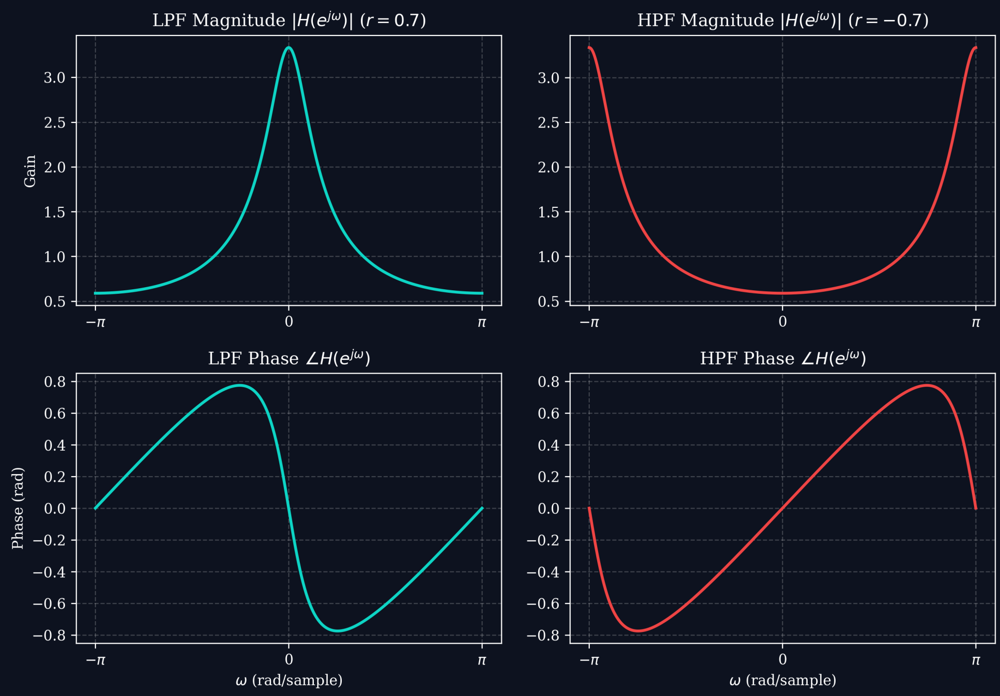


Figure 1: Magnitude and phase responses for first-order digital low-pass and high-pass filters.

# 4 Phase Delay vs. Group Delay

## 4.1 Mathematical Definitions

$$\text{Phase Delay: } \tau_p(\omega) = -\frac{\theta(\omega)}{\omega} \tag{6}$$

$$\text{Group Delay: } \tau_g(\omega) = -\frac{d\theta(\omega)}{d\omega} \tag{7}$$

## 4.2 Narrowband Envelope Propagation

Consider a narrowband wave packet input:

$$x[n] = s[n]\cos(\omega_0 n) \tag{8}$$

Expanding the phase response $\theta(\omega)$ about $\omega_0$:

$$\theta(\omega) \approx \theta(\omega_0) - \tau_g(\omega_0)(\omega - \omega_0) = -\omega_0\tau_p(\omega_0) - \tau_g(\omega_0)(\omega - \omega_0) \tag{9}$$

The resulting system output is:

$$y[n] \approx |H(e^{j\omega_0})| s[n - \tau_g(\omega_0)] \cos(\omega_0(n - \tau_p(\omega_0))) \tag{10}$$

Thus, the envelope $s[n]$ is delayed by the group delay, while the carrier phase is delayed by the phase delay. Figure 2 compares group delay profiles.

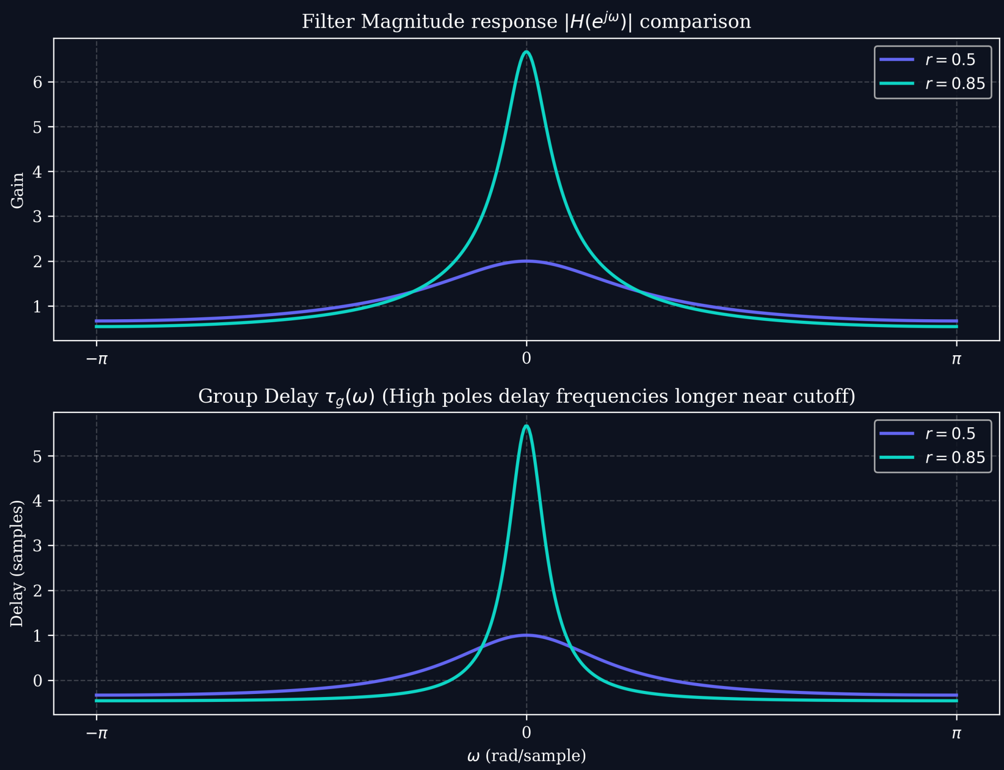


Figure 2: Filter magnitude and corresponding group delay curves for different pole choices.

## 5 Group Delay Derivation for First-Order Filter

Let $y[n] - ry[n-1] = x[n]$ with $|r| < 1$.

$$H(e^{j\omega}) = \frac{1}{1 - re^{-j\omega}} \tag{11}$$

$$\theta(\omega) = -\arctan\left(\frac{r\sin\omega}{1 - r\cos\omega}\right) \tag{12}$$

Let $u(\omega) = \frac{r\sin\omega}{1-r\cos\omega}$. Using the derivative of $\arctan(u)$:

$$\frac{du}{d\omega} = \frac{r\cos\omega - r^2}{(1 - r\cos\omega)^2} \quad \text{and} \quad 1 + u^2 = \frac{1 + r^2 - 2r\cos\omega}{(1 - r\cos\omega)^2} \tag{13}$$

Combining these terms yields:

$$\frac{d\theta(\omega)}{d\omega} = -\frac{1}{1+u^2}\frac{du}{d\omega} = -\frac{r\cos\omega - r^2}{1 + r^2 - 2r\cos\omega} \tag{14}$$

Thus:

$$\tau_g(\omega) = -\frac{d\theta(\omega)}{d\omega} = \frac{r\cos\omega - r^2}{1 + r^2 - 2r\cos\omega} \tag{15}$$